

Computer Networks: Identification Procedures and Communication Types

Summary

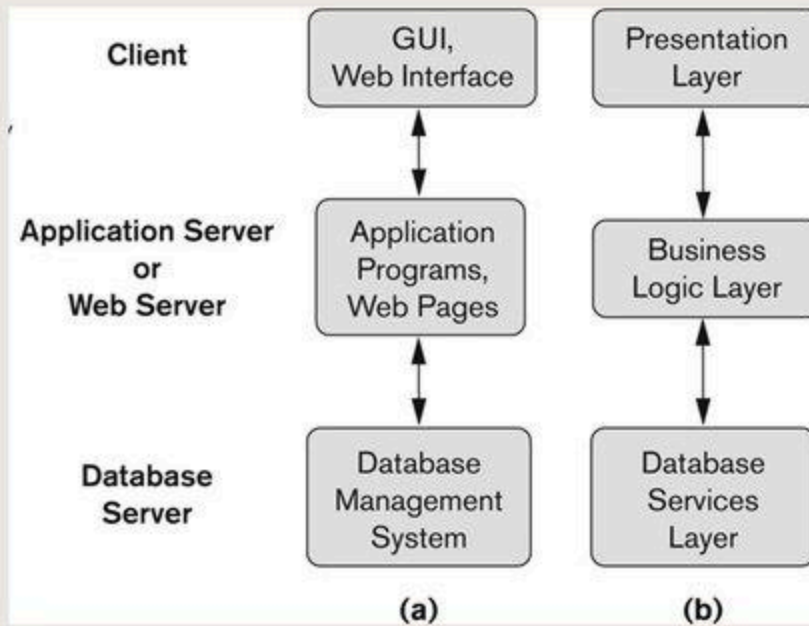
This briefing explains **how data is correctly delivered in computer networks**, focusing on **IPv4 addressing, hierarchical identification, and communication modes**. For any packet to reach the correct destination, networking follows a **three-layer identification hierarchy**:

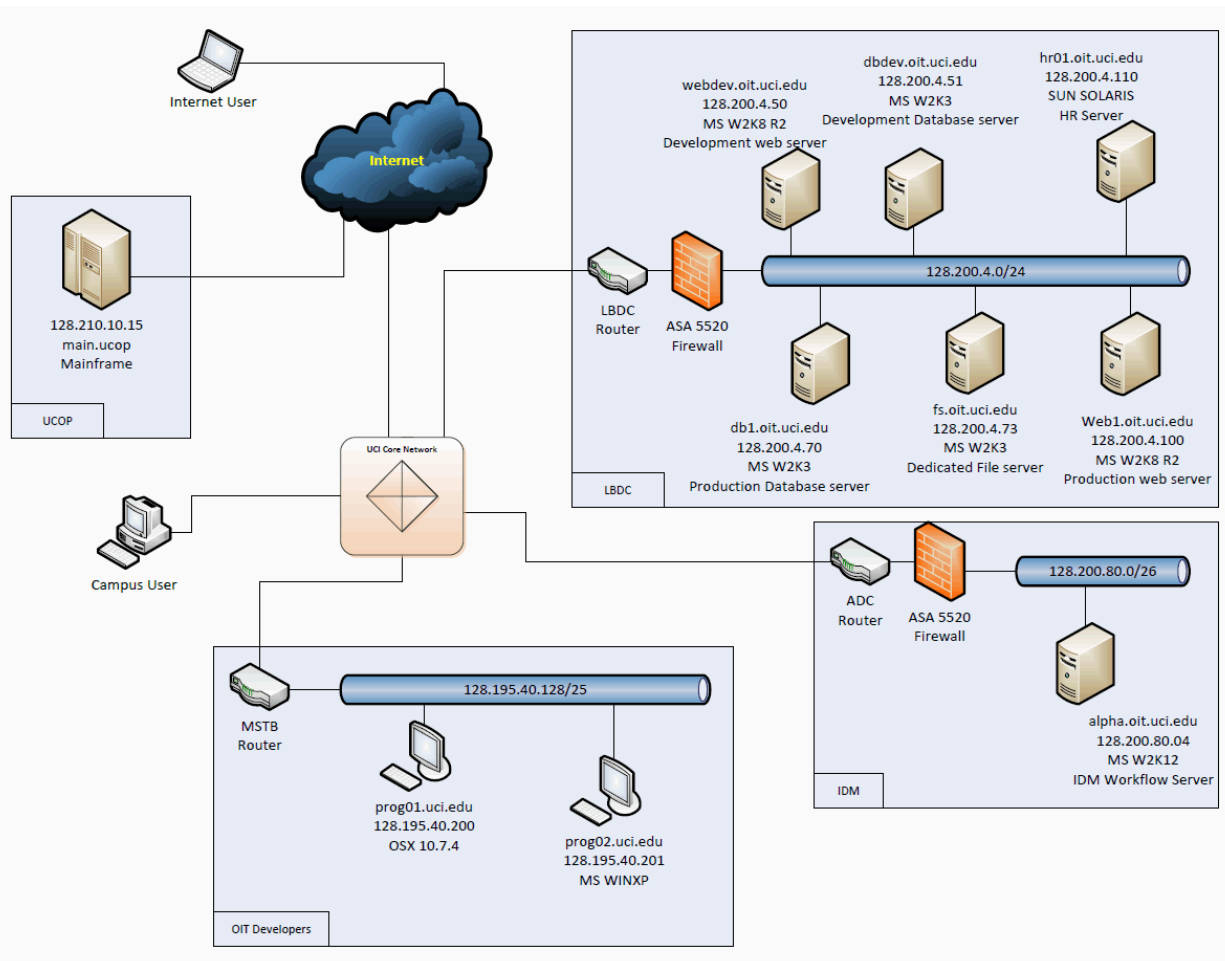
1. **Network identification** (Where is the network?)
2. **Host identification** (Which machine in that network?)
3. **Process identification** (Which application on that machine?)

The document also formalizes **IPv4 host ID rules**, clarifying why **all-zero and all-one host patterns are reserved**, and distinguishes between **Unicast, Limited Broadcast, and Direct Broadcast**, including strict rules about **why broadcast addresses can never act as source IPs**.

The Three-Step Identification Procedure

Three-tier client-server architecture





OSI (Open Source Interconnection) 7 Layer Model

Layer	Application/Example	
Application (7) Serves as the window for users and application processes to access the network services.	End User layer Program that opens what was sent or creates what is to be sent Resource sharing • Remote file access • Remote printer access • Directory services • Network management	
Presentation (6) Formats the data to be presented to the Application layer. It can be viewed as the "Translator" for the network.	Syntax layer encrypt & decrypt (if needed) Character code translation • Data conversion • Data compression • Data encryption • Character Set Translation	E
Session (5) Allows session establishment between processes running on different stations.	Synch & send to ports (logical ports) Session establishment, maintenance and termination • Session support - perform security, name recognition, logging, etc.	L I N
Transport (4) Ensures that messages are delivered error-free, in sequence, and with no losses or duplications.	TCP Host to Host, Flow Control Message segmentation • Message acknowledgement • Message traffic control • Session multiplexing	F I L T E R I N G P A C K E T
Network (3) Controls the operations of the subnet, deciding which physical path the data takes.	Packets ("letter", contains IP address) Routing • Subnet traffic control • Frame fragmentation • Logical-physical address mapping • Subnet usage accounting	
Data Link (2) Provides error-free transfer of data frames from one node to another over the Physical layer.	Frames ("envelopes", contains MAC address) [NIC card — Switch — NIC card] (end to end) Establishes & terminates the logical link between nodes • Frame traffic control • Frame sequencing • Frame acknowledgment • Frame delimiting • Frame error checking • Media access control	S E P
Physical (1) Concerned with the transmission and reception of the unstructured raw bit stream over the physical medium.	Physical structure Cables, hubs, etc. Data Encoding • Physical medium attachment • Transmission technique - Baseband or Broadband • Physical medium transmission Bits & Volts	

IP Address, Port Numbers & MAC Address

IP Address

A unique identifier for a device on network

192.168.1.5

Port Numbers

A virtual point for network communication

HTTP	80
HTTPS	443
SSH	22
SMTP	25

MAC Address

A hardware identifier for a network device

00:1A:2B:3C:4:5E

Successful packet delivery requires **progressive narrowing of scope**, from network → host → process.

1. Identifying the Network (Logical Identification)

The first task is to locate **which network** the destination belongs to.

- **Identifier Used:** IP Address (Logical Address)
- **Rule:**
To obtain the **Network ID**, set **all Host ID bits to 0**

Example (Class C)

IP Address: 192.168.1.10

Network ID: 192.168.1.0

Human-Friendly Resolution

Humans use **domain names** (e.g., unacademy.com), which are translated into IP addresses using **Domain Name System (DNS)**.

2. Identifying the Host (Physical Identification)

Once the network is known, the system must locate the **exact computer** within that network.

- **Identifier Used:** MAC Address (Physical Address)
- **Protocol Used:** Address Resolution Protocol (ARP)

ARP Workflow

1. **ARP Request**
 - Broadcast to all hosts:
"Who has this IP?"
2. **ARP Reply**
 - Unicast from the target host:
"That IP is mine; here is my MAC address."



Key Insight:

IP identifies *where* the host is; MAC identifies *which* device it is.

3. Identifying the Process (Service Identification)

A single host can run **many applications simultaneously**. The packet must reach the **correct process**.

- **Identifier Used:** Port Number (Service Point Address)
- **Port Size:** 16 bits → 0–65535

Well-Known Ports (0–1023)

These ports are standardized and managed by **Internet Assigned Numbers Authority (IANA)**.

Protocol	Port	Service
SMTP	25	Email Transfer
HTTP	80	Web Services
FTP	20, 21	File Transfer
DNS	53	Name Resolution

IPv4 Addressing Rules and Reservations



Address	Function
Network address of all 0s	Interpreted to mean "this network or segment."
Network address of all 1s	Interpreted to mean "all networks."
Network 127.0.0.1	Reserved for loopback tests. Designates the local host and allows that host to send a test packet to itself without generating network traffic.
Host address of all 0s	Interpreted to mean "network address" or any host on specified network.
Host address of all 1s	Interpreted to mean "all hosts" on the specified network; for example, 128.2.255.255 means "all hosts" on network 128.2 (Class B address).
Entire IP address set to all 0s	Used by Cisco routers to designate the default route. Could also mean "any network."
Entire IP address set to all 1s (same as 255.255.255.255)	Broadcast to all hosts on the current network; sometimes called an "all 1s broadcast" or limited broadcast.



IP Address	166.75.229.22 / 16
IP Address	1010 0110 . 0100 1011 . 1110 0101 . 0001 0110
Subnet Mask	1111 1111 . 1111 1111 . 0000 0000 . 0000 0000
Network ID	1010 0110 . 0100 1011 . 0000 0000 . 0000 0000
Broadcast ID	1010 0110 . 0100 1011 . 1111 1111 . 1111 1111
Network ID	166.75.0.0
Network ID	166.75.255.255
Maximum number of hosts = $2^{(32-16)} - 2 = 2^{16} - 2 = 65536 - 2 = 65534$	

Reserved Host ID Patterns

The Host ID portion of an IPv4 address carries special meaning:

1. Network ID (All Zeros)

- Host ID = 000...000
- Represents **the entire network**
- ❌ Cannot be assigned to a host

2. Direct Broadcast Address (All Ones)

- Host ID = 111...111
- Used to reach **all hosts in a specific network**
- ❌ Cannot be assigned to a host

Addressing Structure by Class

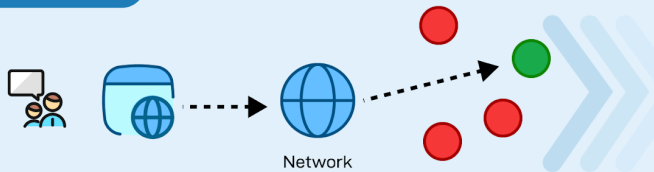
Class	Network Bits	Host Bits	Network Address Format
A	8	24	X.0.0.0
B	16	16	X.X.0.0
C	24	8	X.X.X.0

Types of Communication

Unicast vs Broadcast vs Multicast vs Anycast



Unicast



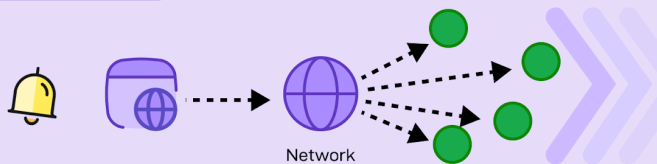
Unique Sender and Single Receiver

Uses a unique destination IP address
Routers look up the routing table to forward



Unicast is used in HTTP, FTP, SMTP

Broadcast



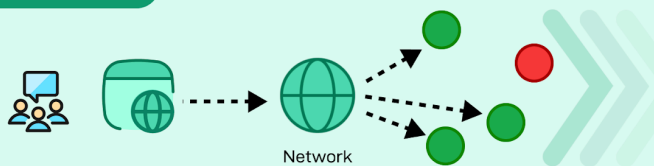
Single Sender and Multiple Receivers

Uses a reserved destination address
Usually restricted to a single logical network



Broadcast is used in ARP, DHCP, NTP

Multicast



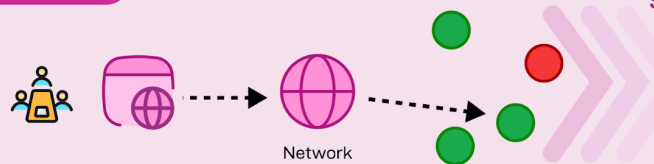
Sender to Specific Receivers

Uses a unique destination IP address
Routers look up the routing table to forward



Multicast is used Video conference apps

Anycast



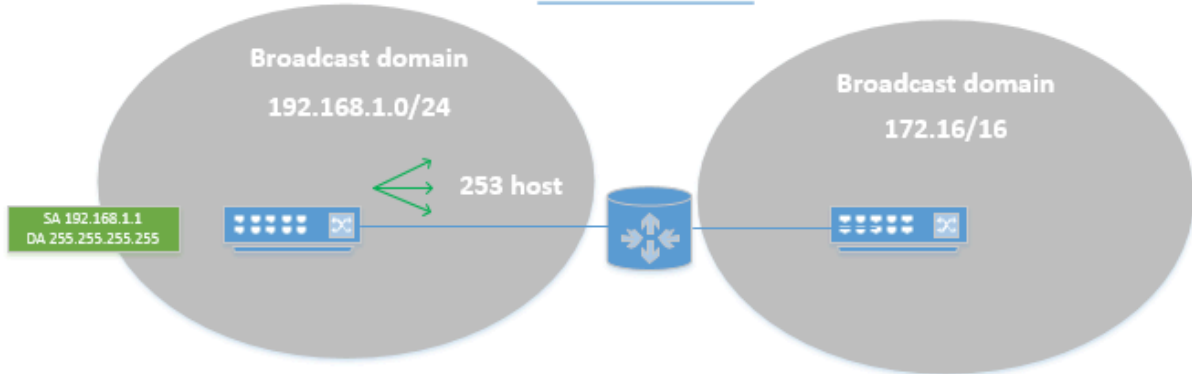
Sender to Single Representative of a Group

Anycast is an extension of unicast
Same unicast address is assigned to multiple devices

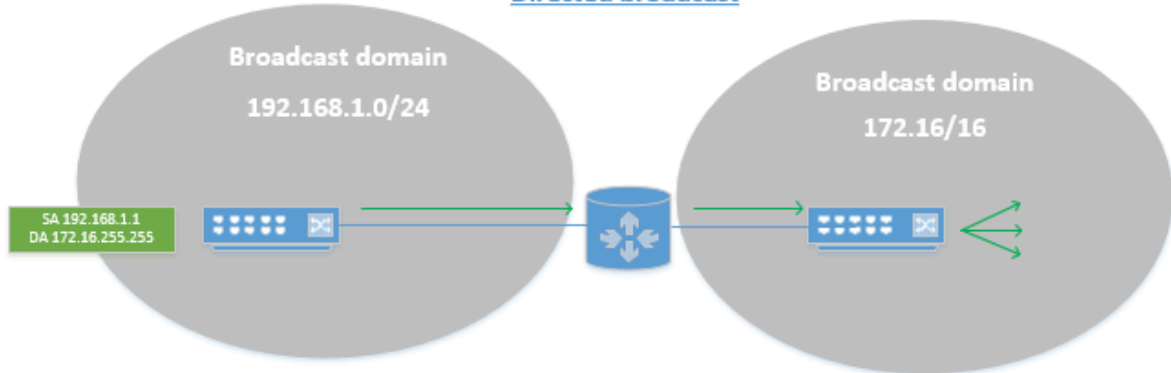


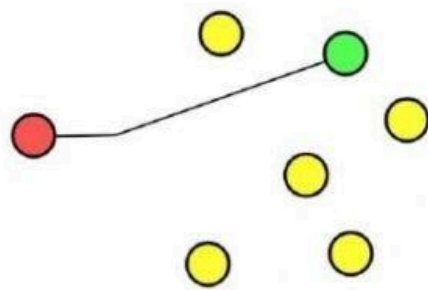
Anycast is used in DNS, CDNs

Limited broadcast

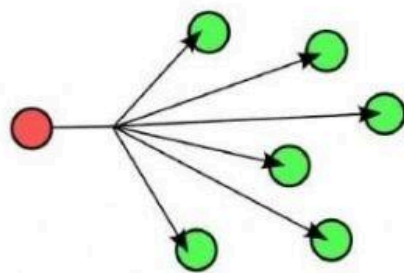


Directed broadcast

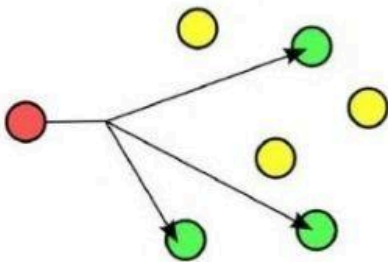




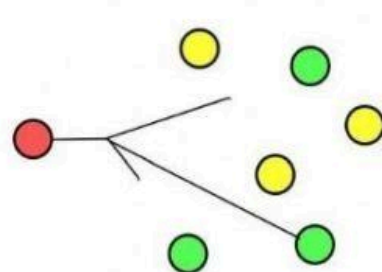
Unicast



Broadcast



Multicast



Anycast

1. Unicast Communication (One-to-One)

- **Definition:** One sender → one receiver
- **Scope:** Same network or different networks
- **IP Validity:**
 - Source IP ✓ valid host IP
 - Destination IP ✓ valid host IP
- **Symmetry:**
 - Source and destination can be swapped

✚ Most internet traffic (web, email, file transfer) is unicast.

2. Limited Broadcast (One-to-All, Same Network)

- **Destination IP:** 255.255.255.255
- **Scope:** Local network only
- **Source Rule:**
 - ✗ Cannot be used as a source IP
- **Class Dependency:**

- Independent of Class A, B, or C

📌 Used by protocols like **ARP** and **DHCP discovery**.

3. Direct Broadcast (One-to-All, Remote Network)

- **Destination IP:**
 - Network ID + all 1s in Host ID

Example (Class A network 11):
11.255.255.255

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- **Source Rule:**
 - ❌ Cannot be used as source IP
- **Reasoning:**
 - A broadcast source would imply **every host sending simultaneously**, which is logically invalid.

Summary of Special IPv4 Addresses

Address Type	Network ID	Host ID	Purpose
Network ID	Valid	All Zeros	Identifies the network
Direct Broadcast	Valid	All Ones	All hosts in a target network
Limited Broadcast	All Ones	All Ones	All hosts in local network
Host IP	Valid	Mixed	Assigned to individual hosts